

1. Administrative & Study Identification

Full Study Title: A Study to Assess the Treatment of Obstructive Hypertrophic Cardiomyopathy (oHCM) With Mavacamten in the US
Short Title/Acronym: MARVEL-HCM (MavAcamten Real World eVidEnce coLLaboration in HCM) **Protocol Number:** CV027-1210 **NCT Number:** NCT07107373 **Posting ID:** 53831077704940842 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Mavacamten (Trade name: Camzyos; ATC Code: C01EB24) **Phase of Development:** Not Applicable **Trial Status:** ACTIVE_NOT_RECRUITING **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the real-world safety and effectiveness of patients with symptomatic obstructive hypertrophic cardiomyopathy (oHCM) initiated on mavacamten at certain high volume HCM centers in the US. **Secondary Objectives:** To describe the long-term changes in functional capacity (NYHA class) and objective echocardiographic parameters (such as Valsalva Left Ventricular Outflow Tract [LVOT] gradient) in a real-world clinical setting.

2.2 Background & Rationale Mavacamten, a first-in-class cardiac myosin inhibitor, is approved for the treatment of adults with symptomatic New York Heart Association (NYHA) class II-III obstructive hypertrophic cardiomyopathy to improve functional capacity and symptoms. However, studies reporting the long-term, real-world effectiveness and safety profile of mavacamten outside of rigorously controlled clinical trials remain limited. This observational study bridges this gap by utilizing electronic medical records (EMR) to evaluate real-world clinical outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective and prospective EMR data collection) **Trial Status:** ACTIVE_NOT_RECRUITING **Control Type:** Single-arm (No control group) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 150 participants
Number of Sites: Multicenter (Including UCSF, Stanford Health Care, Mayo Clinic Arizona, Atlantic Health System, etc.)
Estimated Study Start: December 2023 **Estimated Primary Completion:** 31-Dec-2025

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria**
1. Age ≥ 18 years at the index date.
 2. Prescription of mavacamten for the treatment of Obstructive hypertrophic cardiomyopathy (oHCM) with NYHA Class II or III.
 3. ≥ 12 weeks of follow-up after the initial prescription of mavacamten, except for baseline data reporting.
- 4.2 Exclusion Criteria**
1. Data collection as part of a clinical trial during the study period.
 2. Participation in a myosin inhibitor clinical trial.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen Dosage/Frequency:** Mavacamten (Camzyos) oral capsules dosed and titrated according to standard real-world prescribing labels and the Risk Evaluation and Mitigation Strategy (REMS) program. **Route of Administration:** Oral. **Duration of Treatment:** Chronic, long-term maintenance therapy evaluated over ≥ 12 weeks of follow-up.
- 5.2 Concomitant & Prohibited Therapy Permitted:** Standard of care treatments for oHCM (e.g., beta-blockers, calcium channel blockers) as determined by the treating physician. **Prohibited:** Investigational myosin inhibitors and concurrently enrolling in interventional trials.
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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening / Index	Day 0	EMR review: Prescription date, baseline ECHO (LVOT gradient), LVEF, and NYHA class abstraction.
Follow-up Interval	≥ 12 Weeks	Routine clinical follow-up: Safety evaluation, echocardiography, NYHA functional assessment.
End of Study	Dec 2025	Final data extraction, chart review closure, and overall real-world data synthesis.

(Note: As an observational study, visits align with routine real-world clinical care rather than a strict protocol-mandated schedule).

7. Outcome Measures (Endpoints)

- 7.1 Primary Outcomes**
1. New York Heart Association (NYHA) functional class
 2. Left ventricular outflow tract (LVOT) gradient and associated parameters
 3. Left ventricular ejection fraction (LVEF) and associated parameters

7.2 Secondary Outcomes 1. Mavacamten dose at initiation 2. Mavacamten dose change 3. Reason for mavacamten dose change

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Real-world tracking via clinical notes and EMR abstraction focusing on drug tolerability. **Serious Adverse Events (SAEs):** Captured standardly through routine pharmacovigilance and EMR review. **Data Monitoring Committee (DMC):** Internal sponsor/investigator steering committee reviewing abstracted registry data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled patients fulfilling eligibility with an initial mavacamten prescription and baseline data. **Sample Size Justification:** Descriptive observational cohort; expected target enrollment of 150 based on mavacamten prescription volume at participating high-volume US HCM centers. **Handling of Missing Data:** Analyzed as observed in real-world clinical practice; patients with missing follow-up evaluations for specific parameters are generally excluded from that specific parameter's calculation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and appropriate US Institutional Review Board (IRB) oversight for observational research. **Monitoring Plan:** Periodic quality checks and validation of EMR data abstraction. **Audit Trail:** Secure, de-identified database maintenance mapping directly to abstracted source clinical records.

11. Study Identifiers & Governance

Primary ID: CV027-1210 **Registry Number (NCT):** NCT07107373 **Posting ID:** 53831077704940842 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** MARVEL-HCM **Official Regulatory Title:** MavAcamten Real World eVidENCE coLLaboration in HCM (MARVEL-HCM) **Trial Status:** ACTIVE_NOT_RECRUITING

12. Clinical Framework (oHCM Specific)

Primary Condition: Obstructive Hypertrophic Cardiomyopathy (oHCM) **UMLS Preferred Name:** Hypertrophic Cardiomyopathy (MeSH: D002312 | SNOMED ID: 233873004) **Diagnosis Threshold:** Documented oHCM with

NYHA Class II or III symptoms and objective outflow obstruction (LVOT gradient) **Medical Methodology:** Real-world prescribing of cardiac myosin inhibitors **Optimization Target:** Improvement of functional capacity and reduction of LVOT gradient

13. Study Procedures & Methodology

Management Pathway: Standard clinical management pathways at dedicated, high-volume HCM centers **Education Protocol (HCP):** Mandatory adherence to the Mavacamten REMS program for prescribers **Education Protocol (Patient):** Routine clinical counseling on HCM symptom management and monitoring **Quality Audit Mechanism:** EMR abstraction validation and site-level source data verification

14. Observation & Follow-Up Schedule

Total Duration: Retrospective/Prospective tracking spanning Dec 2023 – Dec 2025 **Onsite Visit Cadence:** Dictated entirely by routine real-world clinical practice (no mandated trial visits) **Remote Monitoring Frequency:** Continuous episodic abstraction of clinical records **Checklist Review Interval:** Minimum of 12-week clinical data checkpoints evaluated retrospectively/prospectively

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				